Tran Trung Duc

AI Engineer | Agentic Systems & LLM Workflows

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PROFILE

AI engineer building Python/FastAPI services and multi-agent LLM systems that run real business operations, from B2B sales and inventory decisions to event, HR and ERP-adjacent document workflows. I own the reliability layer around them: structured outputs, persistent state, retries, validation, human review and evaluation harnesses. My background in biomedical signal processing and QA means I debug what an agent actually did through transcripts, payloads, browser traces and reproducible steps – and keep deterministic calculations in Python while the model handles language.

CORE SKILLS

Backend & cloud	Python (async), FastAPI, REST, SSE, webhooks, Pydantic, SQL/SQLite, JSON Schema, Docker, Cloud Run deployment path, Git, Linux
Agentic systems	Multi-agent orchestration (master/specialist routing), tool/function calling, structured outputs, persistent memory, state machines, retries and fallbacks, human-in-the-loop, MCP, n8n
LLM providers	Anthropic Claude, OpenAI GPT, Llama 3.3 (Groq); prompt and context engineering, structured-output repair, multi-model evaluation
QA & automation	Playwright, Selenium, pytest, contract/integration/E2E tests, regression testing, defect reproduction and severity, evidence capture, transcript debugging, Jira
Product surfaces	HTML, CSS, JavaScript, operator mini-apps, dashboards, review screens, status timelines, postMessage authentication, localStorage fallback
ML & signals	TensorFlow/Keras, PyTorch, scikit-learn, OpenCV, NumPy, Pandas, digital signal processing, model evaluation (MAE/RMSE), TFLite, MIMIC-III

PROFESSIONAL EXPERIENCE

MaxflowAI *AI Engineer / Agent Builder* *Feb 2026 – Present*
Full-time | Ho Chi Minh City, Vietnam

- Built multi-agent systems of up to **7 agents** for sales, operations, HR and inventory workflows: master router, specialist agents, tool calling, persistent memory and structured hand-offs.
- Built FastAPI services with async endpoints, **SSE progress streams** and Pydantic contracts: bounded request fields (1–20 targets, 0–100 scores), typed enums, JSON repair, normalization and invalid-record filtering.
- Cut a 2–3 hour manual research → score → proposal workflow to roughly **3 minutes (>95% reduction)** across 700+ records, with an 11-metric scoring model in deterministic Python.
- Made agent output safe to act on: strict JSON schemas, source-of-truth rules, fallbacks, audit trails and **human confirmation before irreversible sends, posts or exports**.
- Shipped event-driven integrations (webhooks, scheduled jobs, CRM via MCP, SMTP, SMS and messaging), including an event platform with a **12-table schema**, 3 agents, 5 triggers and 3 scheduled jobs; Dockerized services with a Cloud Run deployment path.
- Built **Playwright and pytest harnesses** for auth-aware UI flows, forms, uploads/downloads and contract checks; an ERP receipt workflow passed **88 focused tests** and **5/5 read-back field checks**.

SNOW Vietnam Corporation *Product Reviewer / AI QA Collaborator* *Apr 2025 – Oct 2025*
Ho Chi Minh City, Vietnam

- Designed test scenarios for AI image/video and computer-vision features, targeting low light, occlusion and unusual inputs; filed defects in **Jira** with reproducible steps and severity, giving research a prioritized signal.
- Ran regression checks across model updates and device configurations; partnered with product, engineering and CV research to verify fixes and reduce user-facing defects.

ITR VN *AI Engineer Intern, Deep Learning & Data Pipelines* *Nov 2024 – Jan 2025*
Ho Chi Minh City, Vietnam

- Implemented and evaluated PPG2BP-Net for cuffless blood-pressure estimation (**SBP MAE 9.76±8.99 mmHg**; DBP 5.25±5.26 mmHg; MAP 6.26±5.89 mmHg) on MIMIC-III, from raw waveforms to reproducible evaluation.
- Cut Python data-processing latency by **approximately 40%** with NumPy vectorization; built ingestion, cleaning, feature-extraction and experiment pipelines.
- Prototyped an STM32 + MAX30102 device with Bluetooth streaming to a mobile app for real-time PPG visualization.

SELECTED SYSTEMS

Multi-agent research and lead-scoring service *FastAPI | SSE | Pydantic | Claude tool use*

- Implemented an async tool-use loop with web search and fetch tools, multi-step message context and streamed phase/tool/result/error events to a browser client.
- Enforced 0–100 score bounds, typed enums and JSON contracts; added JSON repair, normalization, dedupe and invalid-record filtering before CRM upsert.
- Kept provisional results explicit when a batch failed validation, so operators could review a weak run instead of trusting it.

Human-reviewed document to ERP posting pipeline *FastAPI | OCR | ERP state machine*

- Modeled the workflow as `draft` → `preflight` → `posting` → `posted/needs_attention`; persisted the posting claim and a correlation ID before the external call, then reconciled on timeout instead of blind retry.
- Added a human-review gate for low-confidence OCR and read-after-write verification; **88 focused tests passed** and **5/5 ERP field checks** matched on BaseType, BaseEntry, item, quantity and warehouse.
- Kept credentials request-scoped and gated production writes server-side; uncertain outcomes became recoverable states with an audit trail and correlation-based reconciliation.

Browser automation runner for agent navigation *Python | Playwright | CLI*

- Built auth-aware browser automation for platform QA: login state, forms, chat, uploads/downloads, deterministic selectors and route discovery when selectors drift.
- Captures screenshots and Playwright traces, runs contract and health checks, and verifies deployed apps end to end; used to reproduce UI defects and confirm fixes.
- Packaged as a CLI with site profiles, read-only checks, dry-run deployment plans and explicit write gates.

AI output verification harness *Python | BM25 | open source*

- Retrieves supporting evidence for each claim from source documents and labels it **SUPPORTED**, **UNSUPPORTED** or **INSUFFICIENT**; handles negation such as “must not” versus “may”.
- Emits re-runnable JSON packs with document IDs, line ranges, confidence and retrieval logs; deterministic verdicts make it usable as a quality gate.
- github.com/BuccKyy/verification-pack-generator

ADDITIONAL PROJECTS

- **Codex plugin (open source)**: turns rough requests into clear, testable, safely scoped work; includes CI checks and an evidence/risk boundary.
- **Deep-PPG-Research-Core**: open-source PPG research framework with signal-quality checks, a model zoo (CNN-LSTM, U-Net, ResNet, Transformer) and evaluation protocols; companion Cuffless-BP-IoT-System covers STM32 acquisition, Bluetooth and a TFLite Android app.
- **Video translation tool**: Whisper ASR → Llama 3.3 translation → Edge TTS → FFmpeg remux, with separate pipeline stages.
- **SGX derivatives downloader**: Python and SQLite; exponential backoff, atomic state tracking and caching that cut repeat runs by 62%.
- **Technical writing**: deployment documentation, test plans, API integration guides and architecture notes that keep debugging and handover self-serve.

EDUCATION & AWARDS

University of Science, VNU-HCM | Ho Chi Minh City, Vietnam *2021 – 2025*
Bachelor of Engineering in Electronics and Telecommunication Engineering | **GPA: 3.4 / 4.0**

Awards: VPBank Technology Hackathon 2025 (Senior Track) | Biomedical Engineering Innovation Competition 2025 (Team Lead)

Certifications: Machine Learning with Python (Coursera) | Deep Learning & Neural Networks with Keras (Coursera) | Building Deep Learning Models with TensorFlow (Coursera)

Languages: Vietnamese (native) | English (professional working proficiency)